
Lecture 9: Logistic Regression

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Classification Problem

Email: Spam / Not Spam?

Online Transactions: Fraudulent (Yes / No)?

Tumor: Malignant / Benign ?

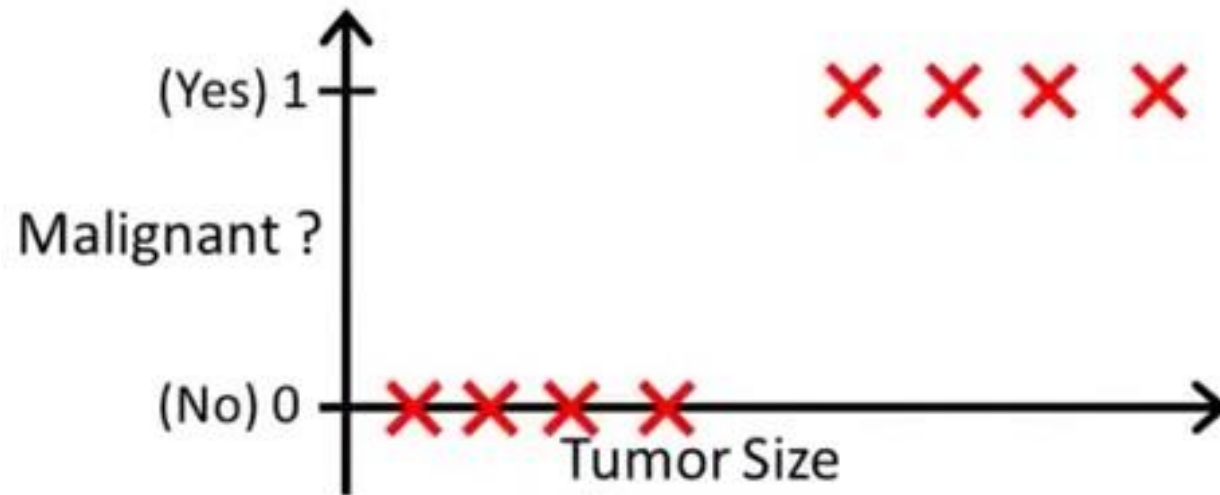
$$y \in \{0, 1\}$$

0: "Negative Class" (e.g., benign tumor)

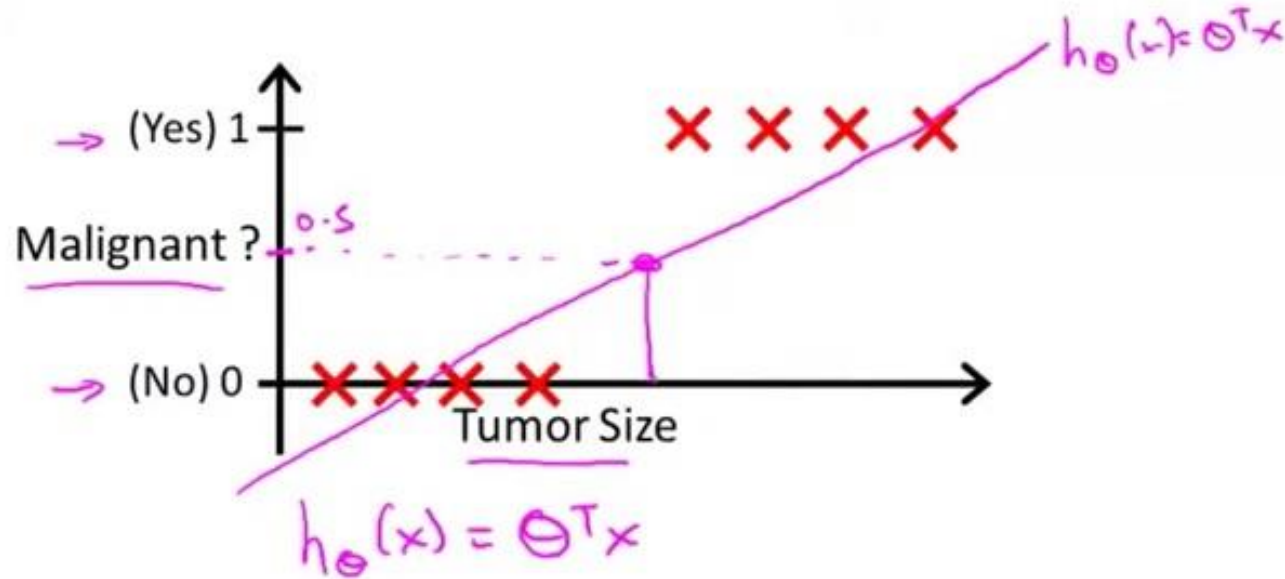
1: "Positive Class" (e.g., malignant tumor)



Solving using Linear Regression



Solving using Linear Regression

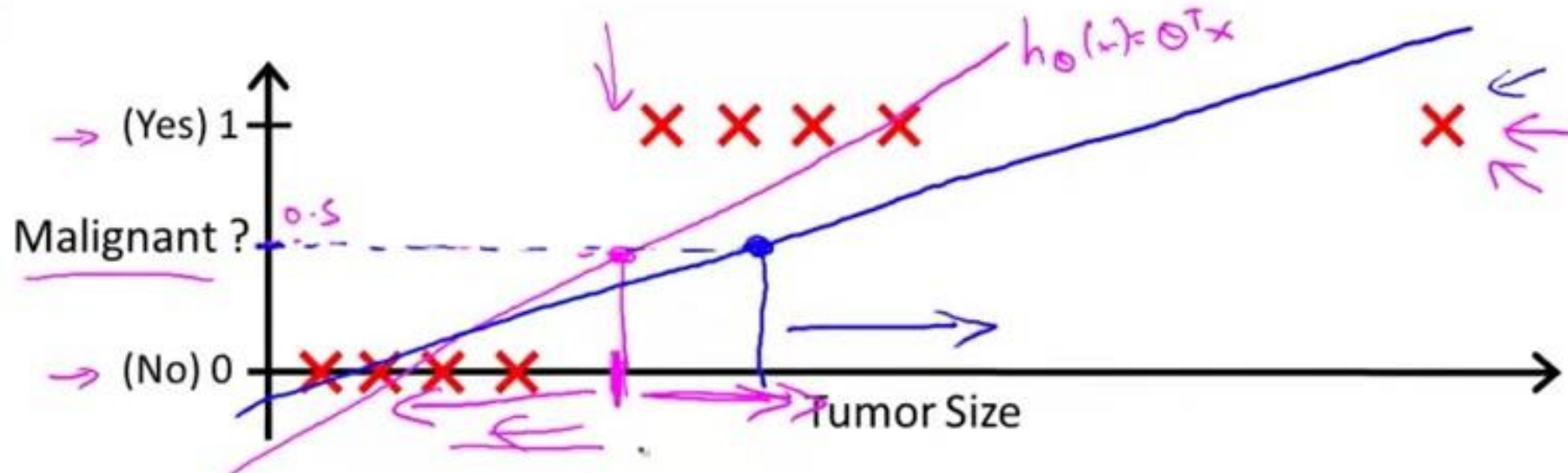


→ Threshold classifier output $h_{\theta}(x)$ at 0.5:

→ If $h_{\theta}(x) \geq 0.5$, predict "y = 1"

If $h_{\theta}(x) < 0.5$, predict "y = 0"

Solving using Linear Regression



Linear Regression vs Logistic Regression

Classification: $y = 0 \text{ or } 1$

$h_{\theta}(x)$ can be > 1 or < 0

Logistic Regression: $0 \leq h_{\theta}(x) \leq 1$

└ Classification

Logistic Regression

Logistic Regression Model

Want $0 \leq h_{\theta}(x) \leq 1$

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

Sigmoid function
Logistic function



Logistic Regression

Logistic Regression Model

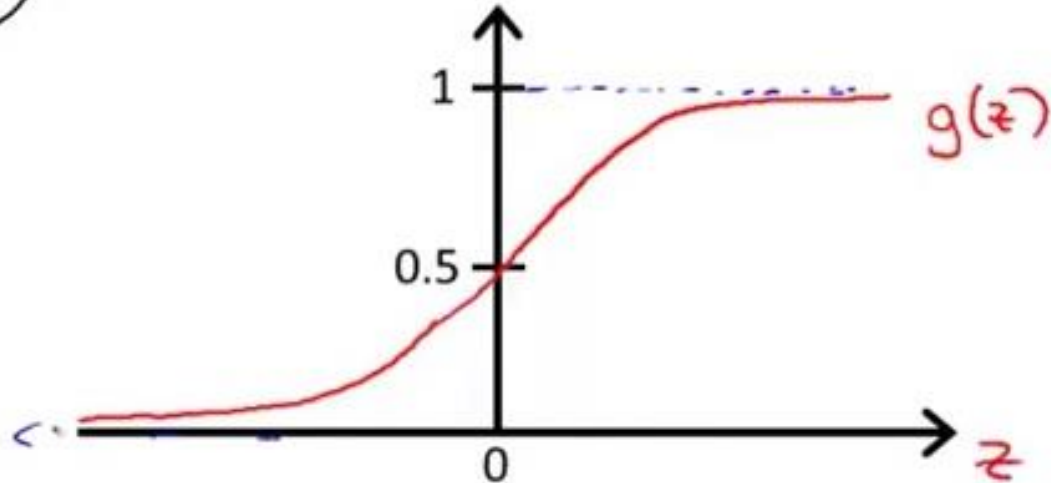
Want $0 \leq h_{\theta}(x) \leq 1$

$$h_{\theta}(x) = g(\theta^T x)$$

$$\rightarrow g(z) = \frac{1}{1 + e^{-z}}$$

$\theta^T x$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$



Sigmoid function
Logistic function

Logistic Regression

Interpretation of Hypothesis Output

$h_{\theta}(x)$

$h_{\theta}(x)$ = estimated probability that $y = 1$ on input x ←

Example: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$$h_{\theta}(x) = 0.7$$

Tell patient that 70% chance of tumor being malignant



Logistic Regression

“probability that $y = 1$, given x ,
parameterized by θ ”

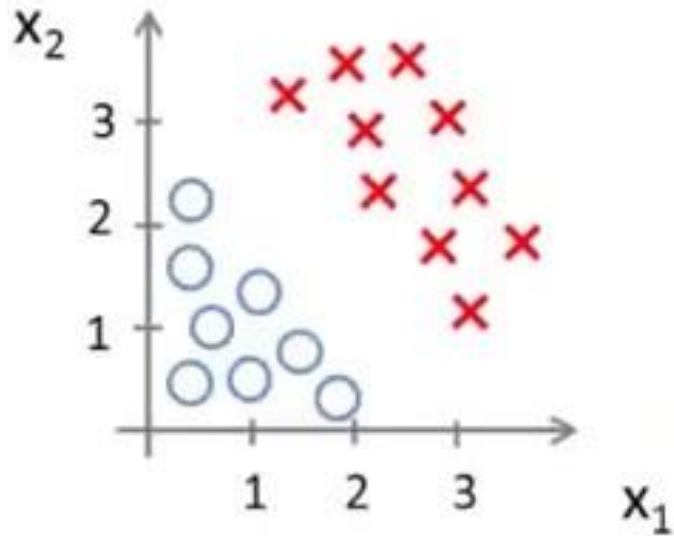
$$P(y = 0|x; \theta) + P(y = 1|x; \theta) = 1$$

$$P(y = 0|x; \theta) = 1 - P(y = 1|x; \theta)$$



Finding Decision Boundary

Decision Boundary



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Predict " $y = 1$ " if $\underbrace{-3 + x_1 + x_2}_{\theta^T x} \geq 0$

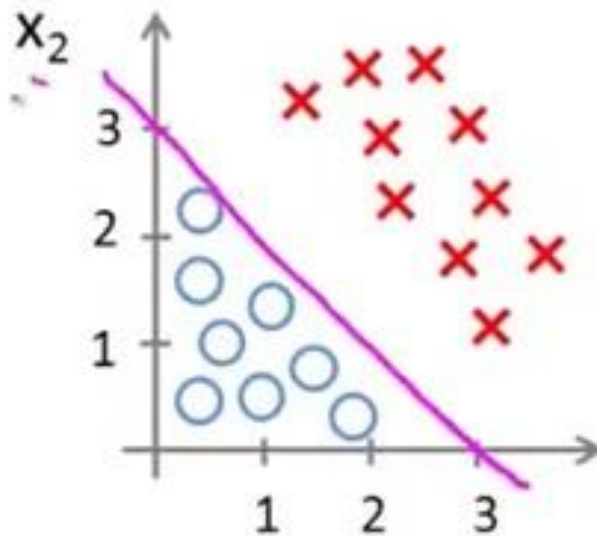
$\xrightarrow{\quad} x_1 + x_2 \geq 3$

x_1, x_2

$x_1 + x_2 = 3$

Finding Decision Boundary

Decision Boundary



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Predict " $y = 1$ " if $\underbrace{-3 + x_1 + x_2}_{\theta^T x} \geq 0$

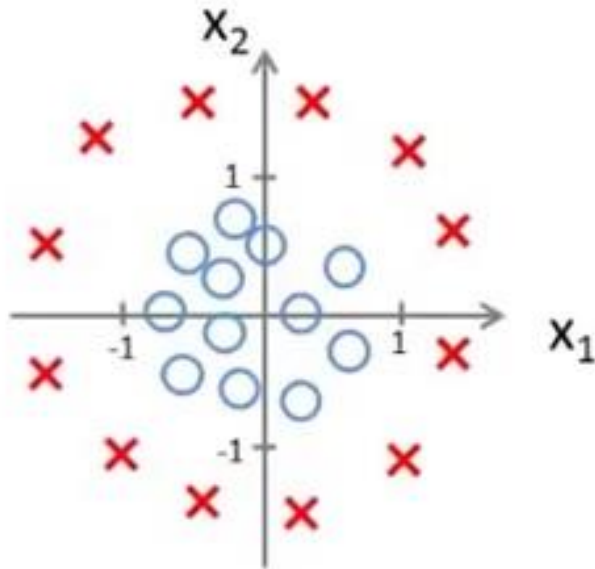
x_1, x_2

$x_1 + x_2 = 3$

$\rightarrow x_1 + x_2 \geq 3$

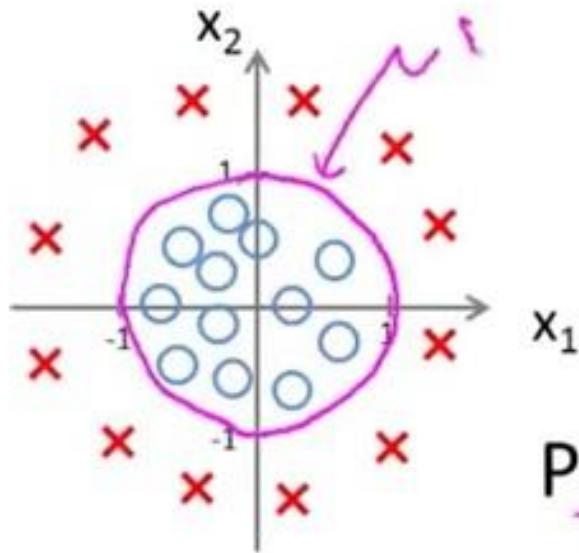
Finding Decision Boundary

Non-linear decision boundaries



Finding Decision Boundary

Non-linear decision boundaries



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

Handwritten annotations above the equation: $\theta_0 = -1$, $\theta_1 = 0$, $\theta_2 = 0$, $\theta_3 = 1$, $\theta_4 = 1$.

$$\theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Predict " $y = 1$ " if $-1 + x_1^2 + x_2^2 \geq 0$

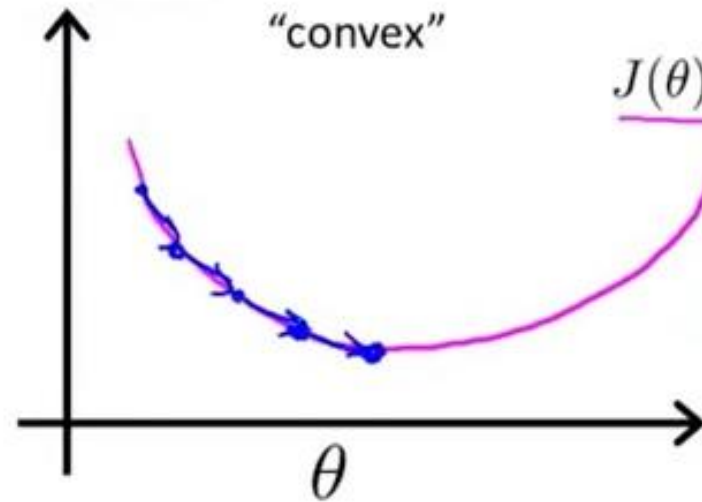
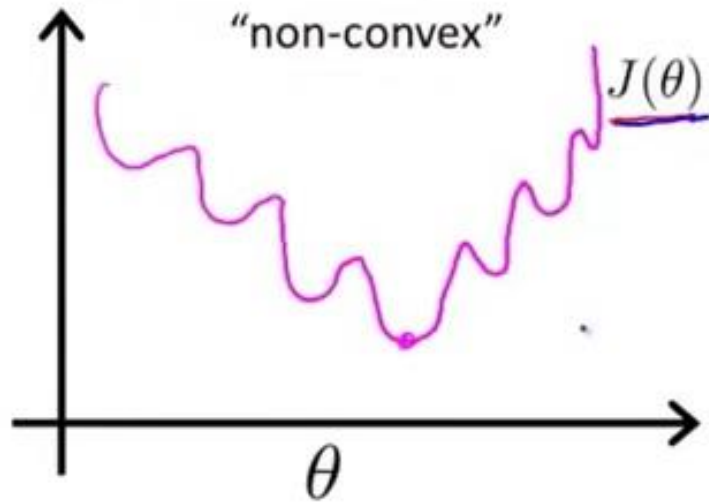
Handwritten notes below the inequality:

- $x_1^2 + x_2^2 = 1$ (circled in purple)
- $x_1^2 + x_2^2 \geq 1$ (underlined in red)

Cost Function for Logistic Regression

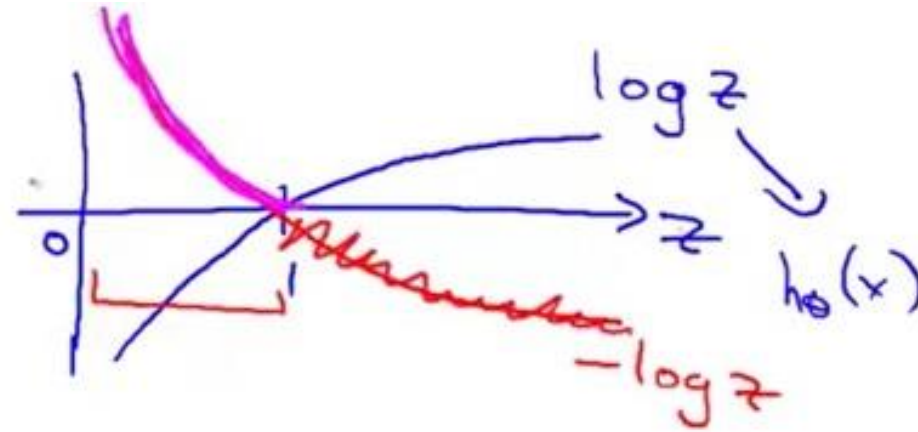
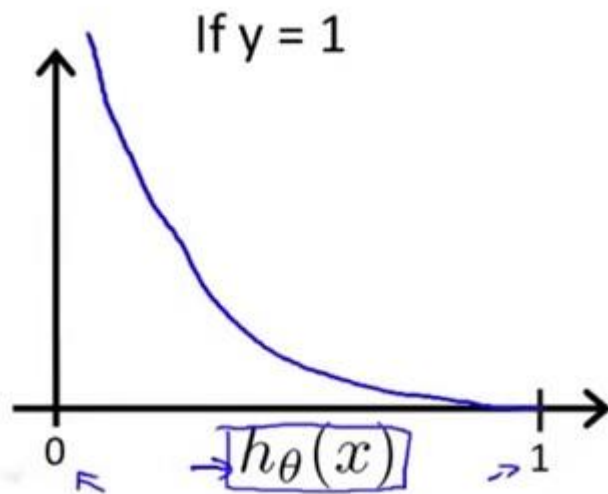
Linear regression: $J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$

$$\text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) = \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

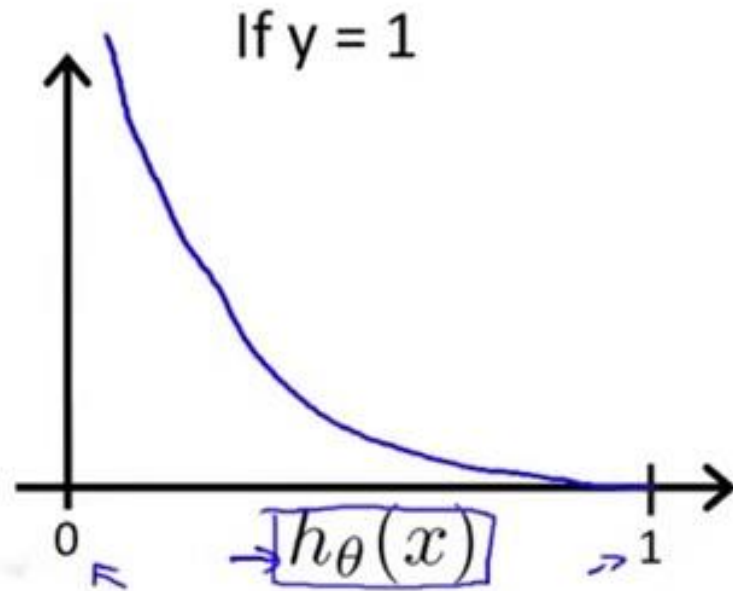


Cost Function for Logistic Regression

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



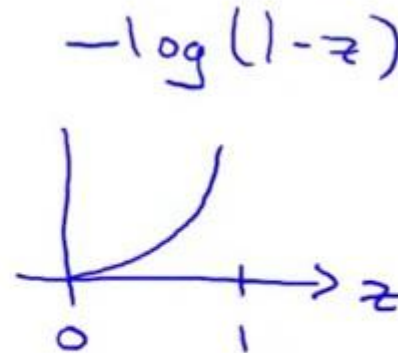
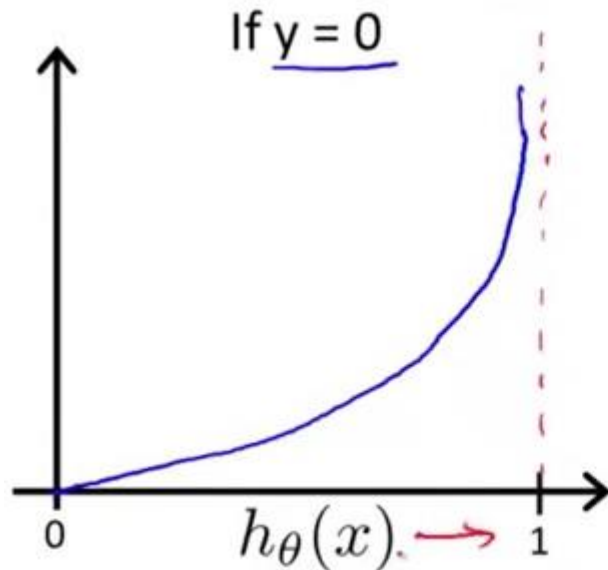
Cost Function for Logistic Regression



Cost = 0 if $y = 1, h_{\theta}(x) = 1$
But as $h_{\theta}(x) \rightarrow 0$
 $Cost \rightarrow \infty$

Cost Function for Logistic Regression

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Cost Function for Logistic Regression

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Note: $y = 0$ or 1 always



Cost Function for Logistic Regression

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Note: $y = 0$ or 1 always

$\rightarrow \text{Cost}(h_{\theta}(x), y) = \underbrace{-y \log(h_{\theta}(x))}_{=0} - \underbrace{(1-y) \log(1-h_{\theta}(x))}_{=1}$

If $y=1$: $\text{Cost}(h_{\theta}(x), y) = -\log h_{\theta}(x) \leftarrow$

If $y=0$: $\text{Cost}(h_{\theta}(x), y) = \underline{-\log(1-h_{\theta}(x))}$

Cost Function for Logistic Regression

Logistic regression cost function

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] \end{aligned}$$

To fit parameters θ :

$$\min_{\theta} J(\theta)$$

To make a prediction given new x :

$$\text{Output } h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$



Gradient Descent

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

} (simultaneously update all θ_j)



Gradient Descent

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

}

(simultaneously update all θ_j)

$$h_{\theta}(x) = \theta^T x$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Algorithm looks identical to linear regression!



Multiclass Classification

Multiclass classification

Email foldering/tagging: Work, Friends, Family, Hobby

$y=1$ $y=2$ $y=3$ $y=4$

Medical diagrams: Not ill, Cold, Flu

$y=1$ 2 3

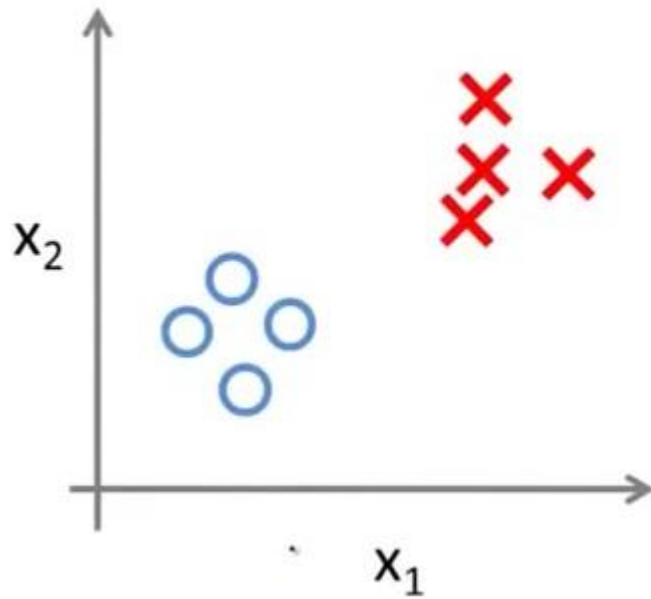
Weather: Sunny, Cloudy, Rain, Snow

$y=1$ 2 3 4

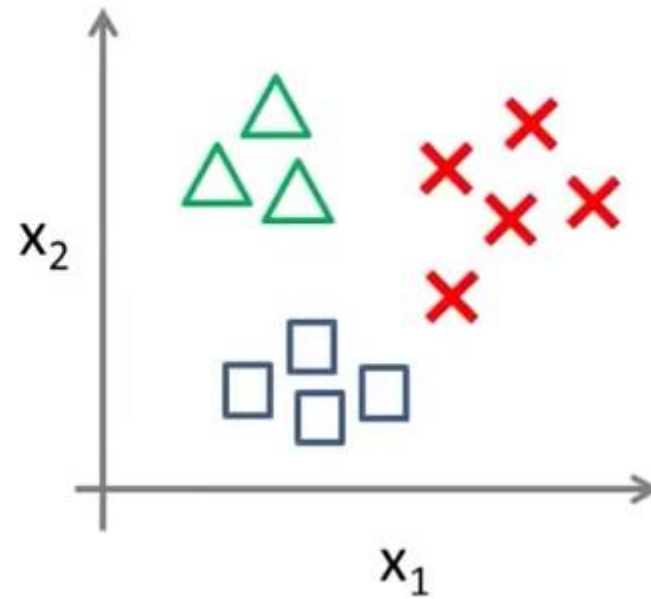


Multiclass Classification

Binary classification:

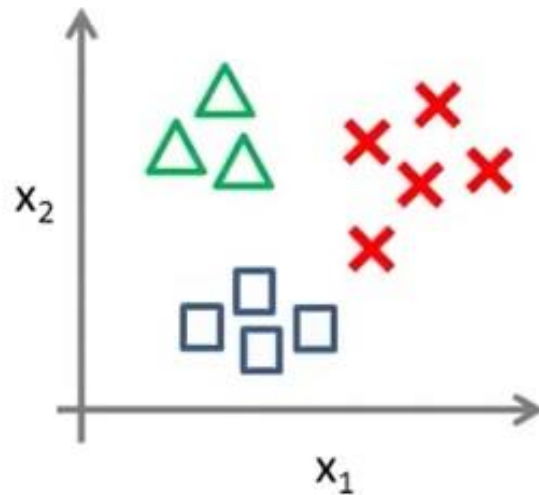


Multi-class classification:



Multiclass Classification

One-vs-all (one-vs-rest):

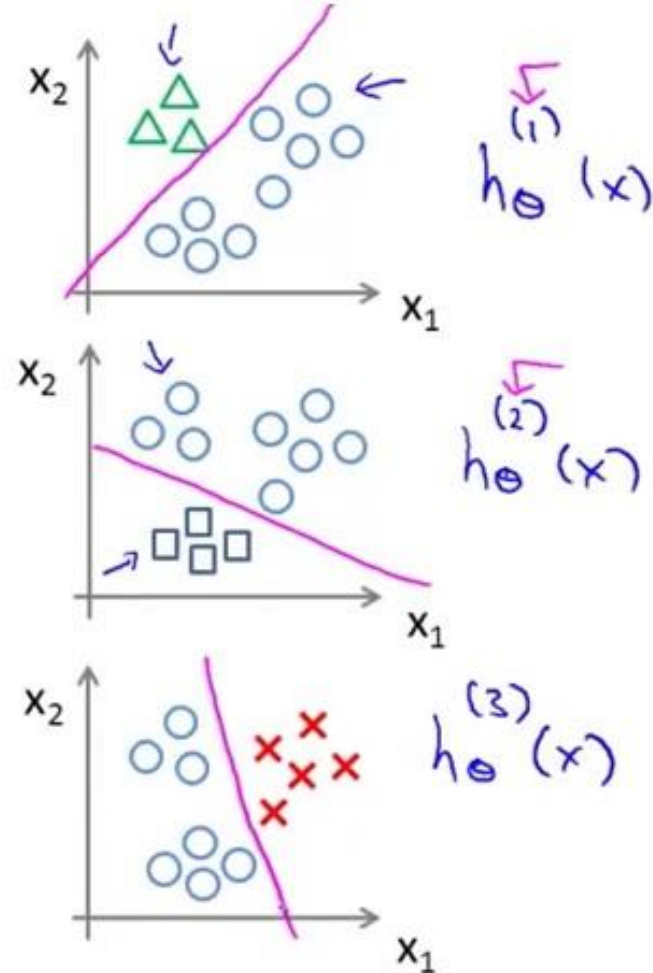


Class 1: $\triangle$

Class 2: $\square$

Class 3: $\times$

$$h_{\theta}^{(i)}(x) = P(y = i|x; \theta) \quad (i = 1, 2, 3)$$



Multiclass Classification

One-vs-all

Train a logistic regression classifier $h_{\theta}^{(i)}(x)$ for each class i to predict the probability that $y = i$.

On a new input x , to make a prediction, pick the class i that maximizes

$$\max_i h_{\theta}^{(i)}(x)$$



Reference

- Andrew Ng Lectures on Machine Learning, Stanford University

